

Md Kaif Imteyaz

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SUMMARY

B.Tech Computer Science student with research experience in machine learning, large language models, and deep learning. Worked with Michigan State University researchers and an Amazon economist on LLM fine-tuning, benchmarking, structured data analysis, and retrieval systems. Experienced in PyTorch, Transformers, model evaluation, machine learning experimentation, and scalable AI pipelines, with ongoing research in reasoning model benchmarking and spatiotemporal forecasting.

EDUCATION

Jamia Hamdard Bachelor of Technology in Computer Science & Engineering;	New Delhi, India 2023 – 2026
Jamia Millia Islamia Diploma in Computer Engineering;	New Delhi, India 2020 – 2023

SKILLS

Programming: Python, Java, JavaScript, C#

Machine Learning & Deep Learning: PyTorch, Scikit-learn, Transformers, OpenCV, Model Evaluation, Experiment Design, Transfer Learning, Fine-Tuning, Model Quantization

LLM & NLP: Hugging Face, OpenAI API, RAG, RLHF, Prompt Engineering, LoRA, QLoRA

Data & MLOps: NumPy, Pandas, Data Preprocessing, Weights & Biases (W&B)

Web & Backend: FastAPI, Node.js, Express.js, React.js, Next.js, Tailwind CSS, REST APIs, Supabase

Databases: PostgreSQL, MySQL

Tools: Git, Jupyter Notebook, Google Colab, Kaggle, Postman, Figma

EXPERIENCE

Data Science Intern Sept. 2025 – Feb. 2026

Worked with Prof. Aaditya Dar, Economist at Amazon

- Implemented a scalable data pipeline that cleaned, standardized, and structured approximately 2 million government official financial asset declaration records for downstream use.
- Designed and executed comparative experiments across BART, FLAN-T5, GPT-2, and Qwen models, evaluating performance trade-offs for structured information extraction tasks.
- Evaluated encoder architectures for entity extraction and record linkage, analyzing trade-offs in accuracy, generalization, and inference efficiency.

Research Intern Mar. 2025 – Aug. 2025

Michigan State University (worked with Prof. Prabhat Barnwal)

- Developed the Peer Knowledge Exchange Platform to improve retrieval speed, accessibility, and user experience for government officers in Punjab, India.
- Deployed a WhatsApp chatbot, Sewa Assist, enabling officers to locate and access program resources faster through conversational retrieval.
- Integrated an on-site chatbot that reduced user friction and improved resource discovery through keyword-driven retrieval logic.
- Delivered the solution using Next.js with version control via GitHub for long-term maintainability and scalability.

Full Stack Developer Intern Dec. 2022 – Jan. 2023

FTK Center for Information Technology, Jamia Millia Islamia

New Delhi, India

- Led full stack development of an alumni community web application serving 2,000+ users, improving engagement using HTML, CSS, JavaScript, and PHP.
- Optimized site architecture for mobile responsiveness and reduced page load times by 20%.
- Collaborated in a four-member Agile team to ship three major features within six weeks.

RESEARCH PROJECTS

Detecting Design Theater in Reasoning Models (Submitted to AIES) | LLM Benchmarking

Developed a benchmark to measure authenticity gaps between generated reasoning traces and actual model behavior through systematic evaluation and interpretability analysis.

Episode-Aware PM2.5 Forecasting Across India | Spatiotemporal Deep Learning

Designed a spatiotemporal deep learning model for country-level PM2.5 forecasting over a 140×124 grid at 25 km resolution, predicting 16 future hours from a 10-hour lookback across 26 input channels with no future meteorological context.

CATS: Culturally Adaptive Therapeutic System | Final Undergraduate Project

Designed an agentic system for physiotherapists to generate personalized exercise videos aligned with patients' cultural backgrounds through culturally appropriate demonstrations.

PROJECTS

Ink2Info: Digitizing Health Informatics | *Python, Next.js, FastAPI, OpenAI, Gemma 7B* | [Repository](#) 2026

- Developed a prescription digitization pipeline using OpenAI and Gemma 7B with a Fabric.js canvas for real-time handwriting extraction, directly modifying inference behavior of the underlying models for domain adaptation.
- Implemented role-based access control with HMAC-signed QR codes for secure data sharing across multiple user roles.

De-Tail: Agentic AI Search Assistant | *React, OpenAI API, Tavily API* | [Repository](#) 2025

- Engineered a retrieval-augmented generation pipeline that exposes its full step-by-step reasoning chain, integrating OpenAI Reasoner with Tavily API for real-time web retrieval.
- Designed end-to-end retrieval flow from query intake through document fetching, reranking signal generation, and response synthesis, with full transparency at each stage.

ACHIEVEMENTS

- **2nd Place**, AMD AI Reinforcement Learning Hackathon among 200+ teams, organized by AMD, Yardi School of AI (IIT Delhi), and Unsloth AI (2026)
- **7th Rank** (out of 2,000+ participants, two-phase), AISEHack 2026 organized by ANRF in collaboration with IBM and IIT Delhi, hosted at IIIT Hyderabad (2026)

CERTIFICATIONS & ACTIVITIES

- **IBM SkillsBuild Certifications:** Artificial Intelligence, Data Fundamentals
- **AI Summer School**, IIIT Delhi (May 2024, July 2023)